

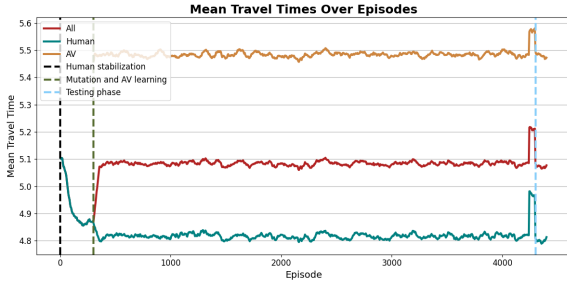
Modifications of IPPO – AIntern Project

Dominika Piechota

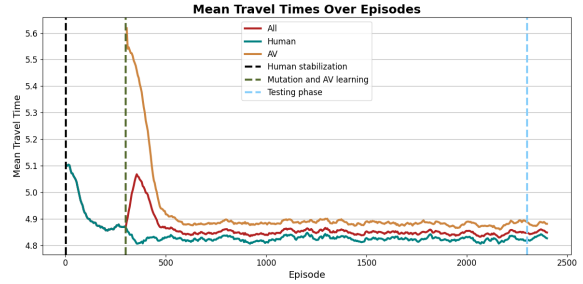
0.1 Centralized Critic and Advantage Estimation

The baseline model relied solely on an actor network mapping raw observations to travel times without a value baseline. A centralized critic network was integrated to evaluate state values, enabling the computation of the Advantage function:

$$A(s, a) = R(s, a) - V(s)$$



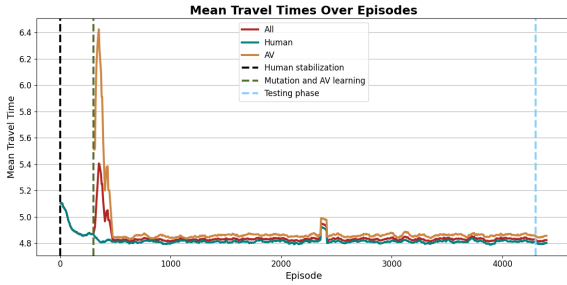
(a) Baseline without critic



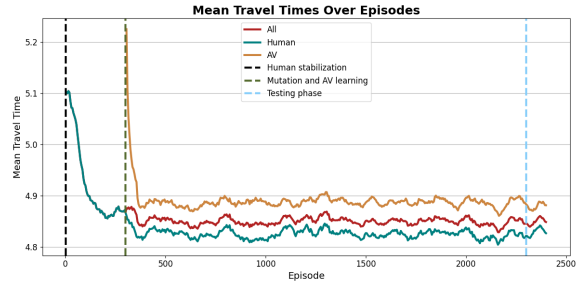
(b) With centralized critic
results/ippo_shared_norm_ing_critic_reference_point

0.2 Different Sampling

To improve sample efficiency, the trajectory sampling mechanism was modified. The baseline algorithm sampled only a single small batch per update and discarded the remaining buffer, wasting the vast majority of simulated experiences. This was replaced with a Mini-batch Gradient Descent approach: the entire episode memory is now randomly shuffled and partitioned into equal mini-batches. Iterating over all batches ensures 100% of the collected data is utilized per epoch before the buffer is cleared. Furthermore, random shuffling breaks temporal correlations, exposing the network to diverse experiences from different locations and simulation stages simultaneously. This accelerates algorithmic convergence.



(a) scripts/ippo_clusters_shared_params_critic
_different_sampling.py

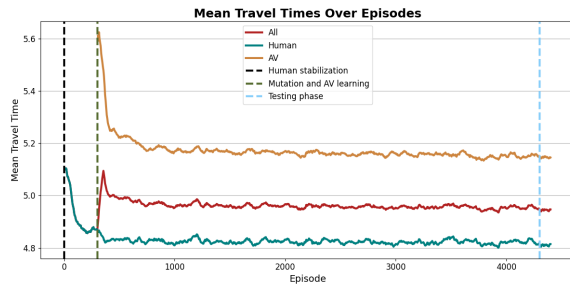


(b) scripts/ippo_clusters_shared_params_critic
_different_sampling.py

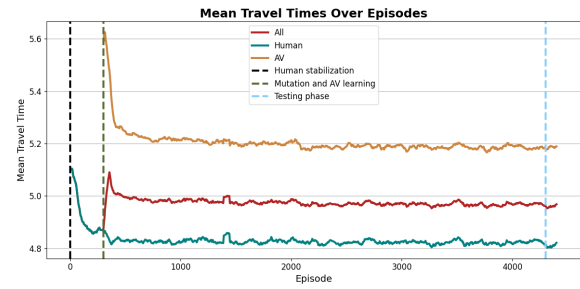
results/ippo_shared_norm_ing_critic_different_sampling results/ippo_shared_norm_ing_critic_different_sampling

0.3 Number of Epochs (num_epoch)

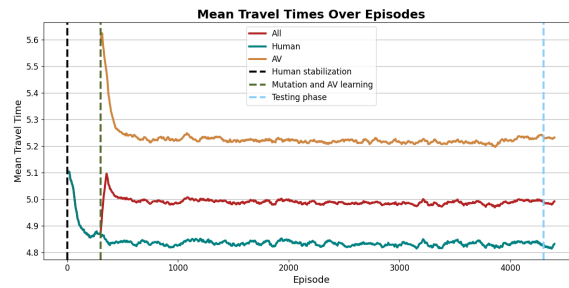
This section presents a brief comparison of the algorithm's performance when modifying the num_epoch parameter.



(a) scripts/ippo_clusters_critic.py
results/ippo_clusters_ing_critic
(Baseline, num_epoch=3)



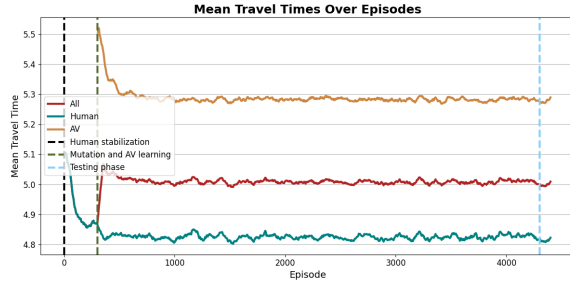
(b) scripts/ippo_clusters_critic.py
results/ippo_clusters_ing_critic_epoch5



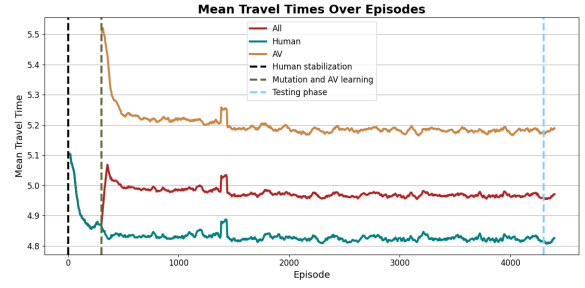
(c) scripts/ippo_clusters_critic.py
results/ippo_clusters_ing_critic_epoch10

0.4 Learning Rate Decay

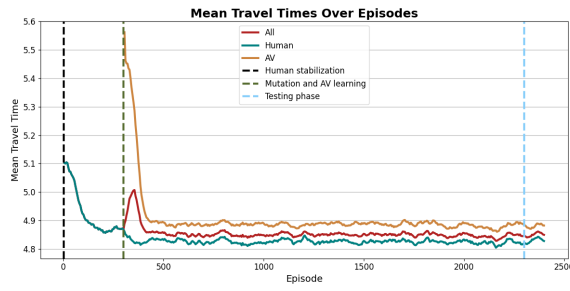
The learning rate decays gradually down to a minimum bound ($lr_{\min}$).



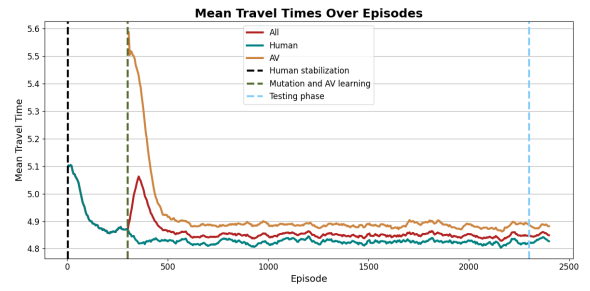
(a) scripts/ippo_clusters_lr_decay.py
results/ippo_clusters_ing_lr_decay1



(b) scripts/ippo_clusters_lr_decay.py
results/ippo_clusters_ing_lr_decay2

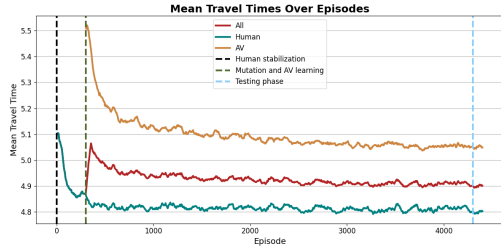


(c) scripts/ippo_clusters_shared_params_critic_lr_decay.py
results/ippo_shared_norm_ing_critic_lr_decay1

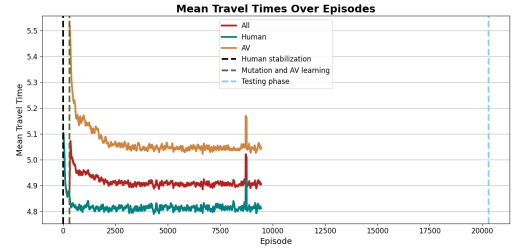


(d) scripts/ippo_clusters_shared_params_critic_lr_decay.py
results/ippo_shared_norm_ing_critic_lr_decay2

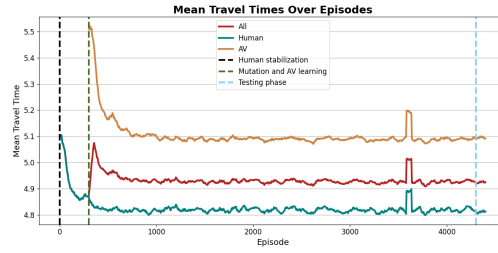
0.5 Entropy Decay



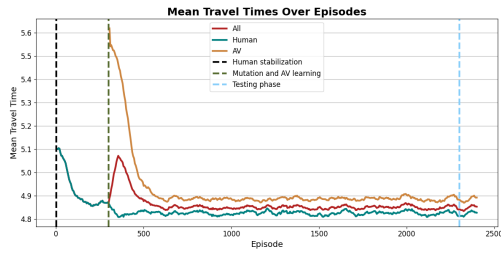
(a) scripts/ippo_clusters_entropy_decay.py
results/ippo_clusters_ing_entropy_decay1



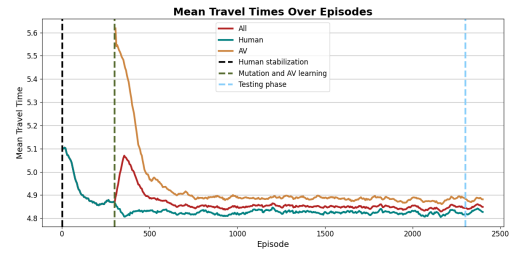
(b) scripts/ippo_clusters_entropy_decay.py
results/ippo_clusters_ing_entropy_decay2



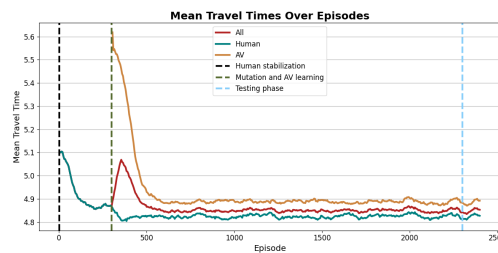
(c) scripts/ippo_clusters_entropy_decay.py
results/ippo_clusters_ing_entropy_decay3



(d) scripts/ippo_clusters_shared_params_critic_entropy_decay.py
results/ippo_shared_norm_ing_critic_entropy_decay1



(e) scripts/ippo_clusters_shared_params_critic_entropy_decay.py
results/ippo_shared_norm_ing_critic_entropy_decay2



(f) scripts/ippo_clusters_shared_params_critic_entropy_decay.py
results/ippo_shared_norm_ing_critic_entropy_decay3

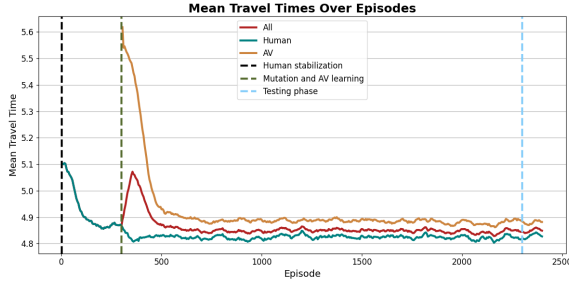
0.6 EMA Reward Centering

An Exponential Moving Average (EMA) baseline was implemented for reward centering:

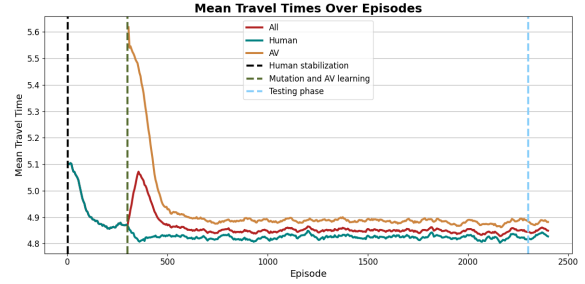
$$\text{ema_baseline} = \alpha \cdot \text{ema_baseline} + (1 - \alpha) \cdot \text{current_mean_reward}$$

$$\text{centered_rewards} = R - \text{ema_baseline}$$

This centers rewards around the current trend, allowing agents to focus on relative performance rather than global traffic fluctuations.



(a) scripts/ippo_clusters_shared_params_critic_ema.py
results/ippo_shared_norm_ing_critic_ema01



(b) scripts/ippo_clusters_shared_params_critic_ema.py
results/ippo_shared_norm_ing_critic_ema005

Figure 6: **Note:** Experiment variants with "norm" in their name utilize normalized observations.

0.7 Overall Conclusions for IPPO

Overall, the majority of the implemented modifications did not yield a significant improvement in the final performance metrics compared to the baseline. However, both *Different Sampling* and the *LR Decay 1* schedule had a notable positive impact on the convergence speed.

